Huawei Access Node KOR-ACC-83 Critical Hardware/Environmental Fault Resolution SOP

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5036

Description

The Access Node KOR-ACC-83, located at the Koramangala Datacenter, utilizes Huawei vendor-specific hardware and software, which is prone to ASIC memory parity errors and line card crashes due to the inherent complexities of its Line Card ASICs. These errors occur when the Line Card ASICs attempt to process high-speed data packets while dealing with the latency and packet loss associated with the optical fiber medium. When the redundancy loss of power supply occurs, it becomes extremely challenging for the system to manage the increased load on the remaining power supply, causing it to crash. The ALARM_5036 indicates a critical failure in the Access Node, necessitating immediate intervention to prevent network downtime. This SOP outlines the necessary diagnostic and resolution steps to rectify the issue, thereby ensuring the stability and high availability of the network.

Pre-requisite Checks

- Ensure that the Koramangala Datacenter is accessible to allow for remote hands to physically inspect and diagnose the issue.

- Verify the redundant link status to ensure that it is functioning as expected.

- Check the environmental conditions of the Koramangala Datacenter, including temperature, humidity, and power supply, to rule out external factors contributing to the failure.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors due to Increased Packet Processing Load	High
Line Card Crashes due to Power Supply Redundancy Loss	Medium
Environmental Factors such as Temperature or Humidity Fluctuations	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-managed fault-info, Expected Output: Fault ID: 5036, Fault Description: ASIC memory parity error, Description: This command will display the fault information, including the Fault ID and description, to help isolate the root cause of the issue.

Command: display device linecard info, Expected Output: Line Card ID: 1, Status: Down, Description: This command will display the line card information, including its status, to help identify if the line card crash is the root cause of the failure.
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Command: display power supply info, Expected Output: Power Supply ID: 1, Status: Down, Description: This command will display the power supply information, including its status, to help identify if the power supply redundancy loss is the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace the Failed Line Card, Command: replace device linecard 1, Description: This command will initiate the line card replacement process. Ensure that the replacement line card is properly installed and configured before proceeding.

Step 2: Step: Replace the Failed Power Supply, Command: replace power supply 1, Description: This command will initiate the power supply replacement process. Ensure that the replacement power supply is properly installed and configured before proceeding.

Step 3: Step: Upgrade the Firmware to the Latest Version, Command: upgrade system firmware, Description: This command will initiate the firmware upgrade process. Ensure that the new firmware version is compatible with the current system configuration before proceeding.

Post-Check Verification

```
Command: display fault-managed fault-info, Expected Output: No faults found, Description: This command will verify that the fault has been cleared and there are no remaining faults in the system.

Command: display device linecard info, Expected Output: Line Card ID: 1, Status: Up, Description: This command will verify that the line card has been successfully replaced and is now functioning as expected.

Command: display power supply info, Expected Output: Power Supply ID: 1, Status: Up, Description: This command will verify that the power supply has been successfully replaced and is now functioning as expected.
```

Roll-back Procedures

- Step: Restore the Original Firmware Version, Command: restore system firmware, Description: This command will restore the original firmware version in case the new firmware upgrade caused the issue.

- Step: Replace the Failed Line Card with the Original Line Card, Command: replace device linecard 1 original, Description: This command will replace the new line card with the original line card in case the new line card caused the issue.

- Step: Replace the Failed Power Supply with the Original Power Supply, Command: replace power supply 1 original, Description: This command will replace the new power supply with the original power supply in case the new power supply caused the issue.